

AI3602 Data Mining Homework 1.2

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Question 1

Frobenius norm error is formalized by $\|A - B\|_F = \sqrt{\sum_{ij} (A_{ij} - B_{ij})^2}$.

Then,

$$\begin{aligned}\|A - B\|_F &= \sqrt{\sum_{i=1}^k U(\sigma_i - s_i)^2 V^T} \\ &= \sqrt{\sum_{i=k+1}^r \sigma_i^2}.\end{aligned}$$

The rest is to prove

$$\min_X \|A - X\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}, X \text{ has rank } k. \quad (1)$$

Use SVD: $X = P\Omega Q^T$, and let $C = P^T A Q$. $A = PCQ^T$.

Write C as

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix},$$

where C_{11} is $r \times r$.

$$\begin{aligned}\|A - X\|_F &= \sqrt{\sum_{i=1}^k (P(\Omega - C)Q^T)^2} \\ &= \|\Omega_{r \times r} - C_{11}\|_F + \|C_{12}\|_F + \|C_{21}\|_F + \|C_{22}\|_F.\end{aligned}$$

There since $X = \arg \min_X \|A - X\|_F$, if C_{12} (or C_{21}) is not $\mathbf{0}$, there exists

$$Y = \begin{bmatrix} C_{11} & C_{12} \\ 0 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} C_{11} & 0 \\ C_{21} & 0 \end{bmatrix}, Y \text{ has rank } k.$$

that makes

$$\|A - Y\|_F = \|\Omega_{r \times r} - C_{11}\|_F + \|C_{21}\|_F + \|C_{22}\|_F \text{ (or } \|\Omega_{r \times r} - C_{11}\|_F + \|C_{12}\|_F + \|C_{22}\|_F) < \|A - X\|_F.$$

This shows $C_{12} = C_{21} = \mathbf{0}$. Similarly, if $C_{11} \neq \Omega_{r \times r}$,

$$Z = \begin{bmatrix} C_{11} & C_{12} \\ 0 & 0 \end{bmatrix}$$

makes $\|A - Z\|_F < \|A - X\|_F$. $C_{11} = \Omega$.

Now $\|A - X\|_F = \|C_{22}\|_F$.

Again use SVD: $C_{22} = P'\Gamma Q^{T'}$. Since C_{11} is diagonal,

$$A = PCQ^T = P \left(\begin{bmatrix} I & 0 \\ 0 & P' \end{bmatrix} \begin{bmatrix} \Omega & 0 \\ 0 & \Gamma \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & Q^{T'} \end{bmatrix} \right) Q^T$$

This is already the SVD form of A , and Ω and Γ each contains some of the eigenvalues of A , since we define X anonymously with $\arg \min$. Anyway, we ensure $\|A - X\|_F \geq \sqrt{\sum_{i=k+1}^r \sigma_i^2}$. This shows equation 1 holds. That's the end of the proof.

Original Questions

Let $A = U\Sigma V^T$ and $B = USV^T$ where $S =$ diagonal $r \times r$ matrix with $s_i = \sigma_i (i = 1 \dots k)$, and $s_i = 0$ otherwise. Please prove B is a best k -rank approximation to A in terms of Frobenius norm error.

Figure 1: Question 1

Reference

Linear Algebra.